

2-Mark Questions & Answers — MPCA Module 4: Memory & Peripherals Interfacing

Q1. What is the role of 74LS138 in memory interfacing?

The **74LS138** is a 3-to-8 line decoder that translates 3 higher address bus lines into 8 active-low Chip Select (/CS) signals to uniquely enable individual RAM/EEPROM memory chips.

Q2. What are the three operating modes of 8255 PPI?

1. **Mode 0 (Basic I/O):** Simple input/output without handshaking.
 2. **Mode 1 (Strobed I/O):** Unidirectional data transfer with Port C handshake signals.
 3. **Mode 2 (Strobed Bidirectional Bus):** Port A acts as an 8-bit bidirectional bus using 5 handshake lines of Port C.
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Q3. Define DMA and list the 4 channels of 8257 DMAC.

DMA (Direct Memory Access) is a high-speed data transfer mechanism directly between I/O peripherals and RAM without CPU intervention.

- The 8257 has **4 independent DMA channels (CH0, CH1, CH2, CH3)**.

Q4. Differentiate between IRR, ISR, and IMR in 8259 PIC.

- **IRR (Interrupt Request Register):** Stores pending interrupt requests.
 - **ISR (In-Service Register):** Tracks the interrupt currently being serviced.
 - **IMR (Interrupt Mask Register):** Stores mask bits to enable/disable specific interrupt lines.
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Q5. What is absolute vs partial address decoding?

- **Absolute Decoding:** All address lines are decoded; no shadow addresses exist.
 - **Partial Decoding:** Some address lines are left unconnected ("don't cares"); creates shadow aliases.
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